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Active Learning-based Isolation Forest (ALIF): Enhancing anomaly detection with expert feedback

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ABSTRACT

The detection of anomalous behaviours is an emerging need in many applications, particularly in contexts where security and reliability are critical. The definition of anomaly varies depending on the domain; however, it is often impractical or too time consuming to obtain a fully labelled dataset. The use of unsupervised models to overcome the lack of labels often fails to catch domain-specific anomalies as they rely on general definitions of outliers. This paper suggests a novel approach to address this problem, Active Learning-based Isolation Forest (ALIF), reducing the number of required labels and tuning the detector to the definition of anomaly provided by the user. The proposed approach is particularly appealing in scenarios where users can interact and provide feedback to the anomaly detector. Smart monitoring software embedded with anomaly detection capabilities commonly relies on unsupervised models, lacking a way to adjust its prediction: ALIF is able to enhance the capabilities of such systems by exploiting user feedback during common operations. ALIF is a lightweight modification of the popular Isolation Forest that proved superior performance compared to other state-of-the-art algorithms in a multitude of real anomaly detection datasets.

1. Introduction

Machine Learning (ML) has gained huge success in the last decades, becoming one of the most popular and studied branches of artificial intelligence. ML methods are widely used in many fields of research, with the aim of obtaining a general working learning rule from input data, namely a prediction function, to be used for future predictions of never-seen-before data. Specifically, ML algorithms have been widely exploited in industrial processes, playing a relevant role in a wide range of applications. Industry 4.0 [1], healthcare [30], transportation [24], and so on.

Anomaly detection represents an important and widely used ML task, broadly applied in various domains and applications where the issue of monitoring unexpected data behaviour is essential. This task defines the process of identifying anomalous data, i.e., data points that are characterised by a different behaviour compared to the rest and that distinguish themselves from the rest of the dataset. To date, there is no unanimously accepted definition, but generally speaking, an anomalous point (often referred to as anomaly or outlier) is defined as *an observation that deviates so much from other observations as to raise suspicion that it was generated by a different mechanism* [13]. Anomaly detection is commonly tackled in many industrial scenarios [40], credit card fraud [4], traffic [16], and medical anomaly detection [9]. In these dynamic and often complex contexts, the problem of detecting anomalies is crucial

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to predict and avoid failures, as well as to perform fault detection. In many industrial processes in fact, data-driven approaches for smart monitoring (for example predictive maintenance) have a key role, allowing to identify and isolate faults and to prevent future sudden failures. To solve this problem, anomaly detection represents an efficient solution. Generally, in this scenario a great amount of collected data is available but, since labelling is an expensive and time-consuming process, there is a lack of ground-truth labels, undoubtedly stating whether or not a point is anomalous. The learning problem therefore is unsupervised, and the algorithm can just blindly look at the structure of the dataset, without a clear definition of what is an anomaly from the user perspective. Therefore, unsupervised algorithms can only detect samples that exhibit some general property different to the rest of the dataset, for example, some approaches look for points far from the majority, or detect points living in low density areas.

Unfortunately, anomalies are highly domain specific [10]: As stated above, since no official definition is given, the concept of what an anomaly is entirely depends on the application in question. Specifically, it may happen that a set of data has different anomalies based on the given application domain and that the same data may be considered anomalous in one domain but normal in another [33]. For example, looking at data acquired by a measuring instrument, the manufacturer might define anomalies as events where the instrument has a faulty behaviour, while the end-user might be more interested in events where the measured process behaves in a previously unseen way [2]. As a direct consequence, training a domain-specific anomaly detector would require a full set of labelled data to capture the user definition of anomaly.

In real-world applications, assigning labels to input data poses a considerable challenge to take into account [41]. In order to train reliable models, a large amount of labelled data is needed, but, in practical scenarios, labelled examples are limited or often too expensive and time-consuming to collect, leading to a huge issue to face. Obtaining labels requires an often too expensive cost to take care of since the labelling procedure is usually carried on by a human domain expert who manually labels each point with a time-consuming and demanding routine. Moreover, by definition, anomalous points are rare and difficult to spot, making the problem a difficult challenge to solve.

Due to the difficulty of finding labelled points, in practical contexts anomaly detection is often treated as an unsupervised learning task. For the classical unsupervised anomaly detection problem, the purpose is to detect outliers with no use of labelled data based on the fact that normal data greatly outnumber anomalous data and anomalies are very different with respect to inliers. Unsupervised anomaly detection models are not optimised for the precise domain of application but are generally based on identifying rules based on specific data characteristics. Density-based methods like LOF [3], COF [39], LoOP [20] and LOCI [31] assume anomalies lie in low-density regions of the domain; distance-based methods like k-NN [19], ABOD [21] and RBRP [11] define anomalies based on some definition of distance from other points; clustering-based methods like OFP [17] and CBOD [18] try to identify points that cannot be easily clustered. Another popular approach are isolation-based methods, such as Isolation Forest (IF) [26,27], which presents a very different approach from most models: Instead of creating a profile for normal data, it explicitly tries to isolate anomalies. To do it, IF relies on two assumptions: anomalies are fewer in number, and they have very different attributes compared to normal data.

Recent literature [6,33] distinguishes outliers from anomalies: The first are the points that simply deviates from the main distribution and are highlighted by an unsupervised model, while the second are the points the user actually sees as anomalous and that can be defined by other domain-specific aspects. As the unsupervised model is not directly tuned to the detection of the anomalies, the outliers might weakly correlate with them.

Therefore, running unsupervised anomaly detection algorithms may be risky and often misleading: as stated previously, anomalies are strongly domain dependent, and as a direct consequence, an unsupervised detector might not identify anomalous data which should be considered as such, and could wrongly flag anomaly points that are normal based on the context taken into consideration [6]. Fig. 1 presents a visual representation of the strong connection between anomalous data points and the context domain. Specifically, the figure shows the two-dimensional projection of the *vowels* dataset [36]. As can be seen, in this case anomalies are not defined just as data points lying far or in low-density regions, but they form a class with a specific pattern defined by domain experts, making it complicated for the automatic detector to correctly identify them. The scenario depicted in Fig. 1 shows how some kind of anomalies might lie in a specific portion of the feature space that could not be identified in any way without some sort of user feedback.

In Decision Support Systems (DSS) [37], data streams are analysed to quickly extract strategic decisions on complex problems. This process is monitored by users who frequently interact with the system and represent the actual decision maker of the whole process. Specifically, if a DSS is present, as a direct consequence, a user is already overseeing the process and inspecting data points [5,28]. Moreover, it has classically been envisioned that user feedback in decision support systems could be exploited in order to improve DSS informativeness; now that DSS are AI-based, this feedback can also be translated in algorithmic improvement of the underlying AD approach. We therefore assume that (i) users/human operators can provide labels to the model and that (ii) the provided labels are accurate.[1] In such a framework, a model that can query the user and take advantage of expert feedback represents an extremely appealing approach.

In this paper, we describe a novel approach, called Active Learning-based Isolation Forest (ALIF), capable of adjusting the detector model on domain specific anomalies by interacting with a human expert. In ALIF, only a subset of data is automatically selected for training: in this way, the number of interactions between the system and the human is minimised. The core idea of ALIF is to iteratively ask labels corresponding to the most ‘significant’ points (with respect to the anomaly detection task), in order to reduce

[1] Assumption (ii) it is reasonable in many settings: in medical domains, feedback can be provided by trained doctors, while in industrial processes, tagging may come from process/equipment experts; in future works, we will relax this assumption and extend the current approach in order to consider a more general framework where users can also provide feedback that are wrong up to a certain degree.

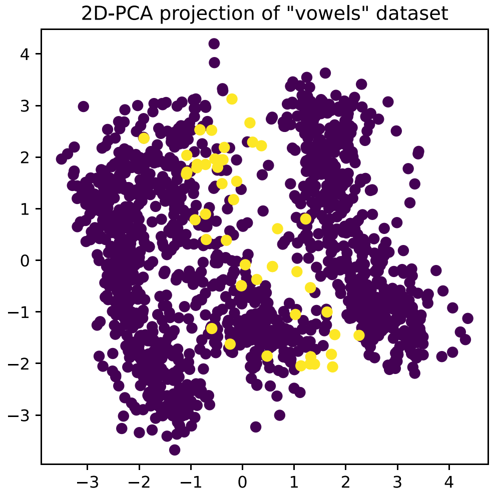

Fig. 1. Projection of the *vowels* dataset on 2 dimensions using Principal Component Analysis. Purple data points are normal data, yellow points are anomalous data. Two aspects are clearly visible: i) it is impossible to separate anomalies from normal data with only two features and ii) anomalies tend to form a different class and might be quite different from general outliers. Not all the data points lying in low density areas or far from the majority are defined as anomalies, but just the ones lying in a specific part of the space. As a consequence, in the considered scenario, identifying anomalies might be a challenging task for an unsupervised detector.

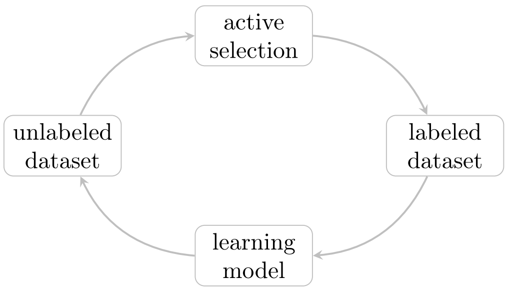

Fig. 2. Active learning core structure. At each iteration a novel point is actively selected from the unlabelled set of data and the corresponding label is requested. Based on the received information, the model is modified.

the labelling cost and, at the same time, to maximise the detection performance. As a direct consequence, the proposed procedure can be seen as an Active Learning (AL) based model [22].

Indeed, AL represents a training approach particularly suitable when labelled samples are too expensive or difficult to obtain. Specifically, AL is a particular ML algorithm based on a key idea: Despite the shortage of labelled data, high-accuracy results may be obtained if the training algorithm is allowed to choose the points to be libelled and learn from them [35]. An AL algorithm asks an oracle to label the data considered most informative with an iterative approach. In doing so, since the queried points are directly selected by the learning algorithm, the amount of necessary labelled data is much smaller than that required for the classical supervised ML approach. Fig. 2 shows the core structure of any AL algorithm: at each iteration the model is updated using the labelled dataset and is allowed to ask for a new label in the unlabelled dataset. This process repeats until the model reaches sufficient performance or when the number of iterations reaches the maximum budget. Moreover, this method has two key advantages over the supervised and computationally expensive models: As it relies on an initial unsupervised training, it can start to work when there are no labels, but more importantly it can work even if instances from only one class are labelled. This is particularly useful when obtaining labels from the anomalous class is very uncommon or expensive.

This work focuses on the Isolation Forest algorithm since it is one of the most effective unsupervised approaches [12], and proposes a strategy to improve the detection performance based on the user's definition of anomaly. The Isolation Forest is highly regarded by scientists and practitioners as one of the most effective approaches in the unsupervised Anomaly Detection field. In this study, two AL query policies for requesting new labels from the user are compared. In addition, two alternative policies for updating the internal structure with minimal computational effort are compared. The goal of the proposed strategies is to enhance the detector performance as much as possible, using minimal labelling effort and an inexpensive model update procedure.

The rest of the paper is organised as follows. Initially, Section 2 outlines the Isolation Forest algorithm and gives a description of an existing AL-based anomaly detection algorithm, which will be used as a benchmark in this work. Then, Section 3 illustrates the proposed approach in detail. In Section 4 ALIF, is compared with other models on a diverse range of datasets used in the Anomaly Detection literature. Finally, in Section 5, present some closing remarks about limitations of the approach and future work direction. The notation used throught the paper is summarized in Table 1.

Table 1
Updated list of symbols used.

Symbol	Description
X	a set of unlabelled training points
x	a single data point in X
F	a forest
T	a tree
L	a leaf of a tree
I_L	number of normal points in L
O_L	amount of anomalies contained in L
$k(L)$	colour of leaf L
$h(L)$	depth of leaf L
$h_T(x)$	depth of the leaf of tree T that contains the data point x
$h^s(L)$	supervised depth of leaf L
$h^s_T(x)$	supervised depth of the leaf of tree T that contains the data point x
$a(x)$	anomaly score for the point x
$a^s(x)$	modified anomaly score for point x

2. Related works

Isolation Forest [26,27] represents an original and innovative anomaly detection algorithm. This algorithm differentiates itself from other existing approaches primarily through its core structure. In the majority of anomaly detection models, normal points play a central role and are used to compute a model of nominal behaviour. Data points that break away from the computed normal instance profile are subsequently labelled as anomalies. In contrast, Isolation Forest constructs its framework around anomalous points. It operates based on the hypothesis that anomalies are generally scarce and possess distinct attribute values, making them more likely to be isolated from the rest of the data by random partitions. Isolation Forest is a non-parametric cost-effective ensemble method that has been found to be successful, often exceeding the results of more elaborate state-of-the-art methods [38]. For unsupervised anomaly detection, it is still one of the best performing approaches [12], while maintaining low train and inference times.

An adaptive anomaly detection algorithm based on IF is presented in [6]. The paper describes an Active Anomaly Discovery (AAD) method in which the points having the highest anomaly score are presented to an expert analyst in an iterative way, requesting the corresponding true label. Based on the information received, the algorithm tries to maximise the anomaly scores assigned to points labelled as anomalies. Specifically [7] presents IF-AAD, where the model starts from the unsupervised IF and retains its tree structure, but also associates to each leaf of each tree a specific weight. These weights are then used to adjust the model output and are updated based on the feedback received. The update procedure is performed by solving an optimisation problem, where the optimal weights ensure that the model assigns a higher anomaly score for points labelled as anomalies than for points labelled as not anomalies.

A radically different approach that is not based on tree ensambles but rather on Support Vector Machines (SVM) was also considered. In particular, [25] describes a semi-supervised algorithm for anomaly detection that attempts to improve the unsupervised model provided by a One Class Support Vector Machine (OCSVM) [32] utilising expert feedback in the form of labelled data points. The proposed model, called ν -SSVM, determines the support vector coefficients by solving a modified version of the optimisation problem used in OCSVM. This formulation is equivalent to OCSVM if no labelled data is provided, and when the whole dataset is labelled, reduces to a standard supervised SVM approach. ν-SSVM handles the trade-off between the unsupervised and supervised approaches in a way that is especially tuned for anomaly detection, considering class imbalances and possible uncertainty in the labelled data. The method can be tuned using two hyperparameters regulating the number of support vectors and miss-classified labelled examples for which the authors provide default values.

In this paper we present a novel approach, called Active Learning-based Isolation Forest (ALIF). Similarly to IF-ADD, our model is equivalent to a standard Isolation Forest, when no labels are provided, but it is able to incorporate user feedback into the model predictions. However, ALIF differs from the aforementioned works because it is specifically designed to be inserted into an active learning loop, relying on local model updates. Unlike IF-AAD, Random Forest and especially ν-SSVM, the proposed approach does not require the solution of an optimisation problem, significantly simplifying the update procedure. In Section 4 we report the performance benchmarking of our approach, IF-AAD, Random Forests and ν-SSVM.

3. Proposed model: ALIF

The proposed approach relies on extending an Isolation Forest model, allowing the possibility to query domain experts for labels. Under this framework, once the base Isolation Forest is fully grown, the algorithm is able to select a data point that should be labelled. The selected point is then presented to a human operator who manually inspects and classifies it. Based on the result of this querying process, the model structure is then modified adjusting its predictions. In order for this process to be efficient, at every iteration the main structure of the Isolation Forest remains untouched, and only the anomaly score calculation is modified. In the following we will analyse each step of ALIF, which can be broken down into three interconnected, yet ultimately independent, tasks:

i. *Unsupervised Fit:* Fitting the baseline Isolation Forest model given an unlabelled data set.
ii. *Update Step:* The process by which a newly acquired label is ultimately used to modify the model prediction.

iii. *Query Step:* The process that determines which point is presented to the domain expert for labelling.

3.1. Unsupervised fit

The first step of ALIF is the training of the base model, a standard Isolation Forest. Similarly to Random Forest [14], Isolation Forest training phase constructs an ensemble of binary decision trees, also known as isolation trees (iTrees). In IF this process is completely unsupervised, taking advantage of the fact that, due to their distinctive characteristics, anomalies are more likely to be isolated using random partitions.

Given an unlabelled training dataset $X \in \mathbb{R}^d$, each iTree is generated from a random sub-sample $X' \subset X$ of size $|X'| = \psi$. The structure of the decision tree is formed through a recursive process, starting from the root node and the entire data subset X'. A decision rule is generated by sampling one of the d attributes at random and a split value uniformly in the range of the observed data. The process continues for the two children nodes independently, where one child uses only the data points where the selected attribute is lower than the split value, while the other child uses the rest. The process stops when a maximum height h_{max} is reached or when a single data point reaches the node. This procedure is very appealing as i) it is independent on the size of the dataset, because of the fixed sample size ψ and ii) it is independent of d, the dimensionality of the dataset. For a given tree T, we say that a data point $x \in \mathbb{R}^d$ is contained in a leaf L, if the path taken by following the decision rules of the tree terminates in L. In this sense, leaf nodes of a given iTree represent a partition of $\mathbb{R}^d$, i.e. each point lies in one and only one leaf in each tree T. Once the training phase is completed, the anomaly score for a point x (i.e. an indicator of the likelihood of anomaly) is computed based on the depth of the leafs that contain it, averaged across all trees in the forest:

$$a(x) = 2^{-\underset{T\in F}{\mathrm{mean}}\left[\frac{h_T(x)}{c(\psi)}\right]} \tag{1}$$

where $h_T(x)$ denotes the depth of the leaf of tree T that contains x, and $c(\psi)$ is a normalisation factor, specifically,

$$c(\psi) = 2H(\psi - 1) - 2(\psi - 1)/\psi$$

and defines the average path length of an unsuccessful search in a binary search tree computed with a set of cardinality ψ.
Based on Eq. (1): when $\underset{T\in F}{\mathrm{mean}}\left[h_T(x)\right] \to \psi - 1$, $a(x) \to 0$ and it is quite safe to consider x a normal point; in contrast, when $\underset{T\in F}{\mathrm{mean}}\left[h_T(x)\right] \to 0$, $a(x) \to 1$ and x are most likely an anomaly; when $\underset{T\in F}{\mathrm{mean}}\left[h_T(x)\right] \to c(\psi)$, that is, when the average path length of x is close to the normalisation factor, then $a(x) \approx 0.5$ and the sample does not have a recognisable anomalous factor. The anomaly score can be converted into a classification by selecting a threshold value. The determination of the optimal value for this threshold is intricately linked to the specific characteristics of the data at hand, making the task of selecting an appropriate value quite complex in practice [15].

3.2. Update step

Once the initial unsupervised fit is completed, the proposed iterative active learning approach can start. At each iteration, a point $x \in X$ is selected, the corresponding true label is requested, and the model is modified accordingly. In order to make the IF model updatable, we modify the definition of the anomaly score, making it dependent on the acquired labels. To do so, the depth $h(L)$ used in the IF anomaly score is replaced by a supervised value $h^s(L)$, which depends on the labelled data contained in L.

$$a^s(x) = 2^{-\underset{T\in F}{\mathrm{mean}}\left[\frac{h_T^s(x)}{c(\psi)}\right]} \tag{2}$$

The Algorithm 2 summarises the procedure described above.

Following the rationale of IF, leafs that contain anomalies should be located near the root of the tree, while leafs that contain normal points should be far from the root, we define the *colour* of a leaf L based on the proportion of anomalies and normal points contained in it:

$$\bar{k}(L) = \frac{O_L - I_L}{O_L + I_L}, \tag{3}$$

where I_L and O_L are respectively the number of labelled normal points and the number of labelled anomalies contained in the leaf. Therefore, the colour of a leaf defines its inner structure and describes how the total amount of labelled data contained in it is distributed. Specifically, it outlines the probability that a leaf is anomalous with respect to the labelled data in it. Fig. 3 shows how the *colour* of a leaf is calculated. For convenience, let us define a rescaled version of the colour function, so that its codomain is given by the closed interval [0, 1]:

$$k(L) = \frac{1}{2}\left(\bar{k}(L) + 1\right). \tag{4}$$

Specifically, the following observations can be made:

- If a leaf L contains no labelled points, $k(L)$ is undefined.
- If L contains only normal points then $k(L) \to 0$;

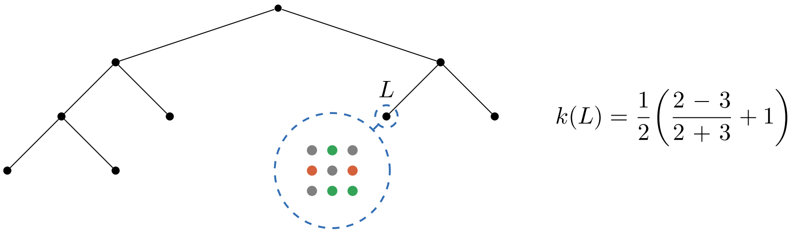

Fig. 3. Every time a novel point is queried and the corresponding label is obtained, every iTree is investigated and the leaves containing the queried point are considered. A visual representation of the colour of a generic leaf L is displayed: green dots represent normal labelled instances; labelled anomalies are depicted by red dots. The corresponding colour $k(L)$ is computed using Equation (4).

- If L contains only anomalies then $k(L) \to 1$;
- If L has an equal number of anomalies and normal points then $k(L) = \frac{1}{2}$.

The supervised depth $h^s(L)$ can then be defined based on the leaf colour. We define two different approaches to compute $h^s(L)$:

1. *Logarithmic* supervised depth

$$h^s(L) = -c(\psi)\log_2(k(L)). \tag{5}$$

2. *Piece-wise Linear* supervised depth

$$h^s(L) = \begin{cases} 2k(L)\left[c(\psi) - h_{max}\right] + h_{max}, & \text{if } 0 \le k(L) < \frac{1}{2} \\ 2k(L)\left[h_{min} - c(\psi)\right] + 2c(\psi) - h_{min}, & \text{if } \frac{1}{2} \le k(L) \le 1 \end{cases} \tag{6}$$

where h_{max} and h_{min} are, respectively, the minimum depth and the maximum depth of the Isolation Forest computed during the unsupervised training.

The logarithmic depth in Equation (5) is a very natural way to map the colour, taking values in the compact interval [0, 1], to the positive reals. It is defined so that:

- when $k(L) \to 1$, then $h^s(L) \to 0$;
- when $k(L) = \frac{1}{2}$, then $h^s(L) = c(\psi)$;
- when $k(L) \to 0$, then $h^s(L) \to +\infty$.

Unfortunately, the vertical asymptote makes the logarithmic choice quite unstable when normal points are labelled. However, it is possible to apply a threshold on the logarithm to saturate it, leading to a behaviour very similar to the piecewise linear function.

The piecewise linear approach is introduced to address this instability. It restricts the domain of $h^s(L)$ to the range of depths empirically observed in the base IF, and then fits a piecewise linear function so that:

- when $k(L) \to 1$, then $h^s(L) \to h_{min}$;
- when $k(L) = \frac{1}{2}$, then $h^s(L) = c(\psi)$;
- when $k(L) \to 0$, then $h^s(L) \to h_{max}$.

Fig. 4 plots the relation connecting the leaf value $k(L)$ with the supervised depth $h^s(L)$ for both the piecewise linear depth and the logarithmic depth. It is important to note that the proposed update strategy does not require the forest to be fully re-trained, but, in contrast, when new information is acquired, ALIF only modifies the colour of actively sampled leaves, leaving the rest unchanged. This can be performed efficiently without much overhead by keeping track of I_L and O_L as leaf properties. Therefore, the time complexity of the update strategy is linear with respect to the number n_T of trees in the forest. Like the original Isolation Forest, this approach is by design independent of the number of features.

Algorithm 1: *ALIF* (F, X).

Data: unlabelled dataset X^u, forest F
Result: updated forest F
$x^q \leftarrow GetQueryPoint(F, X)$;
$y^q \leftarrow GetLabel(x^q)$;
$F \leftarrow LeafDepthUpdate(F, x^q, y^q)$;

3.3. Query step

The idea of incorporating expert feedback in unsupervised anomaly detection algorithms aims at improving performance, adding a relatively small computational and labelling cost. To improve the model in an optimal manner, the choice of the appropriate

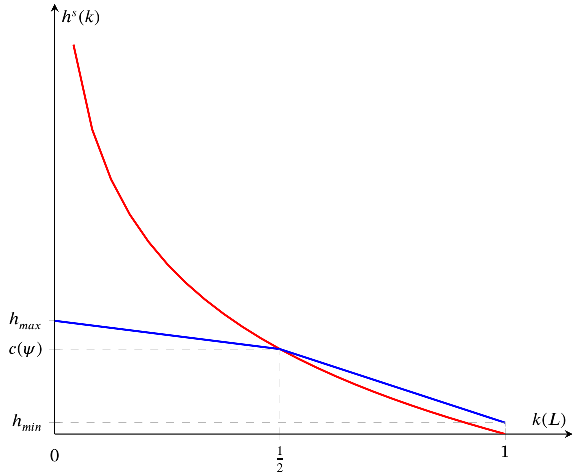

Fig. 4. Visual representation of the two possible definitions of supervised depth. The blue function represents the piece-wise linear depth: when the leaf value $k(L)$ is close to 0, i.e., the leaf is fully labelled with normal points, the novel supervised depth of L is tended to the maximum depth; if $k(L)$ takes a value close to 1, the leaf will be push to the minimum depth, as this situation corresponds to L fully labelled of anomalies. The red function shows the logarithmic depth: in the range extremes 0 and 1, a threshold value is applied and the depth values are forcibly set to respectively h_{max} and h_{min}.

Algorithm 2: *LeafDepthUpdate*(F, x^q, y^q).

```
Data: labelled point (x^q, y^q), unlabelled dataset X, forest F
Result: updated leaf L
for T in F do
    L ← GetLeafNode(F, x^q) ;
    if y^q == anomaly then
        O_L ← O_L + 1 ;
    end
    else
        I_L ← I_L + 1 ;
    end
end
```

strategy used to select points to be queried must be established. In fact, beyond the strategy used to modify the structure of the Isolation Forest based on the novel information, the update performance is significantly dependent on the choice of the queried points. Specifically, for the successful outcome of the method, choosing the most appropriate point to be labelled is a key factor which could compromise its outcome. In classical active learning scenarios, several possible query strategies are presented [34].

Consider an unlabelled dataset X of candidate points. This could be the data set on which the IF has been trained or a different batch of (for instance data collected during deployment). We define the two following query strategies, where the choice of the point to request is based entirely on the supervised depth h^s_T:

1. *Maximum uncertainty*: at each iteration the point selected to be labelled x^q is the one where the iTrees disagree the most, namely

$$x^q = \underset{x \in X}{\operatorname{argmax}} \left[\underset{T \in F}{\operatorname{std}} \left[h^s_T(x) \right] \right] \tag{7}$$

   By doing so, the intended purpose would be to provide greater assistance to the model. Specifically, at each iteration, we ask for the label of the data point where the model is most uncertain, adding information with respect to the most uncertain data.
2. *Most anomalous*: at each iteration, the point selected to be labelled x^q is the point that has the highest anomaly score value in the current iteration, namely

$$x^q = \underset{x \in X}{\operatorname{argmax}} \left[\underset{T \in F}{\operatorname{mean}} \left[h^s_T(x) \right] \right] \tag{8}$$

   This query strategy is the less expensive and more straightforward one and involves asking for the label of the point regarded as the most anomalous.

Note that in order to make the proposed approach feasible, for each of the listed strategies, each point may only be queried once. When a point is queried and the corresponding label is obtained in fact, there is no need to ask for its label again, since we are considering the information received undoubtedly true.

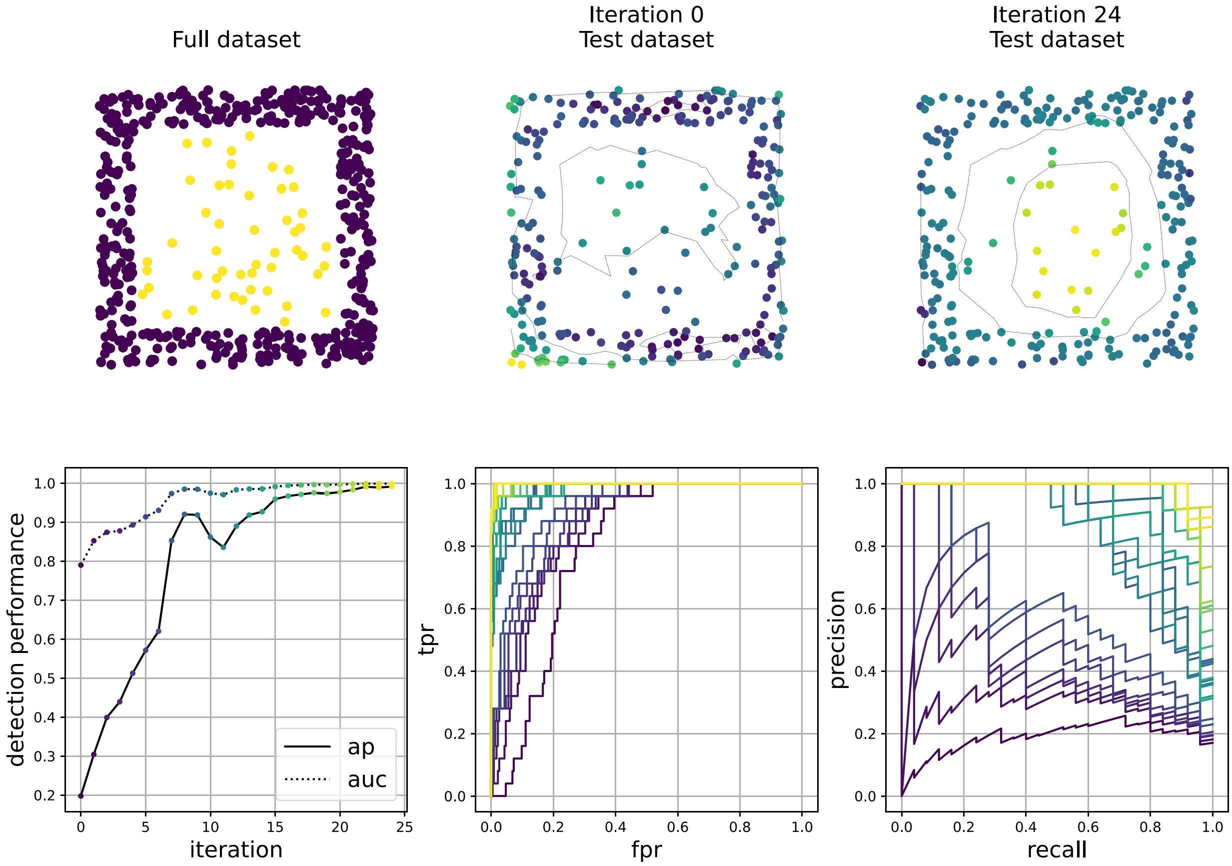

Fig. 5. Test of the algorithm on a square toroidal dataset: anomalies lie inside of a box made up of normal instances. This setting is particularly challenging for the Isolation Forest algorithm as it is much easier to quickly separate normal points (depicted in purple) w.r.t. anomalies (yellow). The anomaly score on test samples is shown on the second and third panel: the yellow is assigned to the points having the highest anomaly score, the purple viceversa. In the second row of panels the performances of the detector at each iteration are depicted: the *area under the ROC curve* (auc) and the *average precision* (ap) measured on the test set quickly improve.

From a business process point of view, querying the most anomalous point may represent a sort of DSS, providing assistance to the decision making process and at the same time extracting information from a significant amount of data. As stated above, a DSS software gathers data considered the most informative and, based on the information achieved, it generates analysis tools useful for the decision-making process.

In this scenario, when a point is flagged as a strong anomaly by the system, it triggers the involvement of a domain expert for further scrutiny. Through this process, novel insights are obtained, combining the expert's knowledge together with the computerised system. Collaboration between the expert and the system produces improved decision-making validation, while also rapidly extracting valuable information to enrich the decision-making process.

From a model-based viewpoint, asking for the point where the iTrees are most uncertain may be the most relevant strategy. With this approach, in fact, the assistance provided by the expert aims at addressing the decisions where the model struggles the most and, doing so, at updating the most critical situations. Of course, given the unbalanced number of anomalies, it is highly likely that, using the maximum uncertainty strategy, the vast majority of the labelled points would be normal data, making it possibly less suitable in a more practical perspective.

As specified by Equations (7) and (8), ALIF query strategy need to retrieve the depth of all leaves for each datapoint in X. This procedure has the same computational complexity as performing inference for all candidate data points in X and is independent of the number of features.

4. Experimental results

In this section, we analyse the performance of ALIF, for all combination of proposed update strategies and query approaches. Doing so, we hope to obtain a full and detailed evaluation of the presented model, with the purpose of analysing the efficiency of each combination as well as giving meaningful guidance on the proper use of the proposed strategies.

Firstly, we tested our approach on synthetic datasets like the challenging shape depicted in Fig. 5, where the normal data make up a square toroid and the anomalous data lie inside it. In this kind of datasets, the Isolation Forest performs quite poorly and there

Table 2
Set of data used in the experimental phase. The first column gives the name of the dataset; the second column describes the number of instances contained in each set; the third column defines the total amount of features; the fourth column gives the number of outliers; the last column presents the contamination rates.

Dataset	Instances	Features	Anomalies	Contamination
AnnThyroid	7200	6	534	7.42%
Breastw	683	9	239	35%
Cardio	1831	21	176	9.6%
Cover	286048	10	2747	0.9%
Ionosphere	351	33	126	36%
Letter	1600	32	100	6.25%
Mammography	11183	6	260	2.32%
Mnist	7603	100	700	9.2%
Optdigits	5216	64	150	3%
Pendigits	6870	16	156	2.27%
Pima	768	8	268	35%
Satellite	6435	36	2036	32%
Satimage-2	5803	36	71	1.2%
Thyroid	3772	6	93	2.5%
Vertebral	240	6	30	12.5%
Vowels	1456	12	50	3.4%
WBC	278	30	21	5.6%
Wine	129	13	10	7.7%

is a lot of room for improvement as it struggles to separate in few steps the anomalies: in this case it is much easier to wrongly separate normal points w.r.t. anomalies, indeed the first iteration of the model corresponding to the unsupervised training gives the highest anomaly score to the normal bottom left point of the toroid. On the contrary, at the 25-*th* iteration the model has learnt the correct function and is able to perfectly classify all the points. This is also visible in the panels of the second row of Fig. 5, where the performance of the detector at each iteration is depicted with lines having different colours. As the model is allowed to query new points, the average detection performance quickly improves.

We decided to test the model on a set of 18 real data openly available [36,8]. Table 2 presents the dataset used to test the performance of the proposed model. These come from different domains like medicine, industry and natural sciences and are characterised by various numbers of points as well as percentage of anomalous data. Defining the contamination as the ratio between the number of anomalies and total number of samples of the dataset, they are characterised by a wide feature range between 6 to 100 and a contamination percentage between 0.9% and 36%. The datasets were selected specifically for being built in the following way: they are adaptations of multi-class classification datasets to the anomaly detection task where one of the classes is under sampled and labelled as outlier. This process produces points considered outliers that are not just randomly scattered in the feature space like general anomalies, but they lie in specific, structured regions of the space that are usually hard to identify without labels. For this reason, this type of anomalies is usually quite difficult to identify using only unsupervised approaches. Therefore in this context, weakly supervised methods can prove their efficacy, starting from an unsupervised guess, and continuously improving as labels are included in the model. This indeed allows to take advantage of both the unsupervised and supervised approaches, getting a model that is highly flexible.

We used the average precision score metric to measure the results obtained. Splitting the data set equally into training and testing sets, we performed a number of 25 queries, and 50 experiments were carried out independently to study the performance distribution. We limited the maximum number of queries to 25 for two reasons: i) in scenarios where labelling is costly, a higher number of queries is typically unfeasible; ii) for larger number of queries we may be dealing with problems where supervised approaches could be naturally applied. The experiments were performed on equipment with an Intel Core i7-6800K CPU and 32 GB RAM. The implementation of ALIF is freely available online.[2]

First, we tested the four possible combinations of the two proposed update strategies together with the two query strategies in order to determine the best combination with respect to the majority of the test sets considered. Fig. 6 shows the results obtained.

Given that the first point of the curve represents the performance of the fully unsupervised Isolation Forest, it can be observed as, broadly speaking, all four proposed matches represent an improvement with respect to the performance of the Isolation Forest. The two trials of the “most anomalous” query are depicted with a solid line, while the “maximum uncertainty” is depicted in a dashed line.

It is clear that no strategy is strictly better than the others, however, the piecewise linear update strategy is the one having the most stable improvements among the datasets. This result is in line with the remarks on the instability of the logarithmic update made in Section 3. In most cases, the most anomalous query policy seems to have the fastest improvements. This is a very interesting aspect since the “most anomalous” strategy is the cheapest and most natural policy among the two considering the DSS scenario previously described. In fact, points that have high anomaly scores would raise an allarm in the DSS and would be investigated regardless. In

[2] https://github.com/tombarba/ActiveLearningIsolationForest.

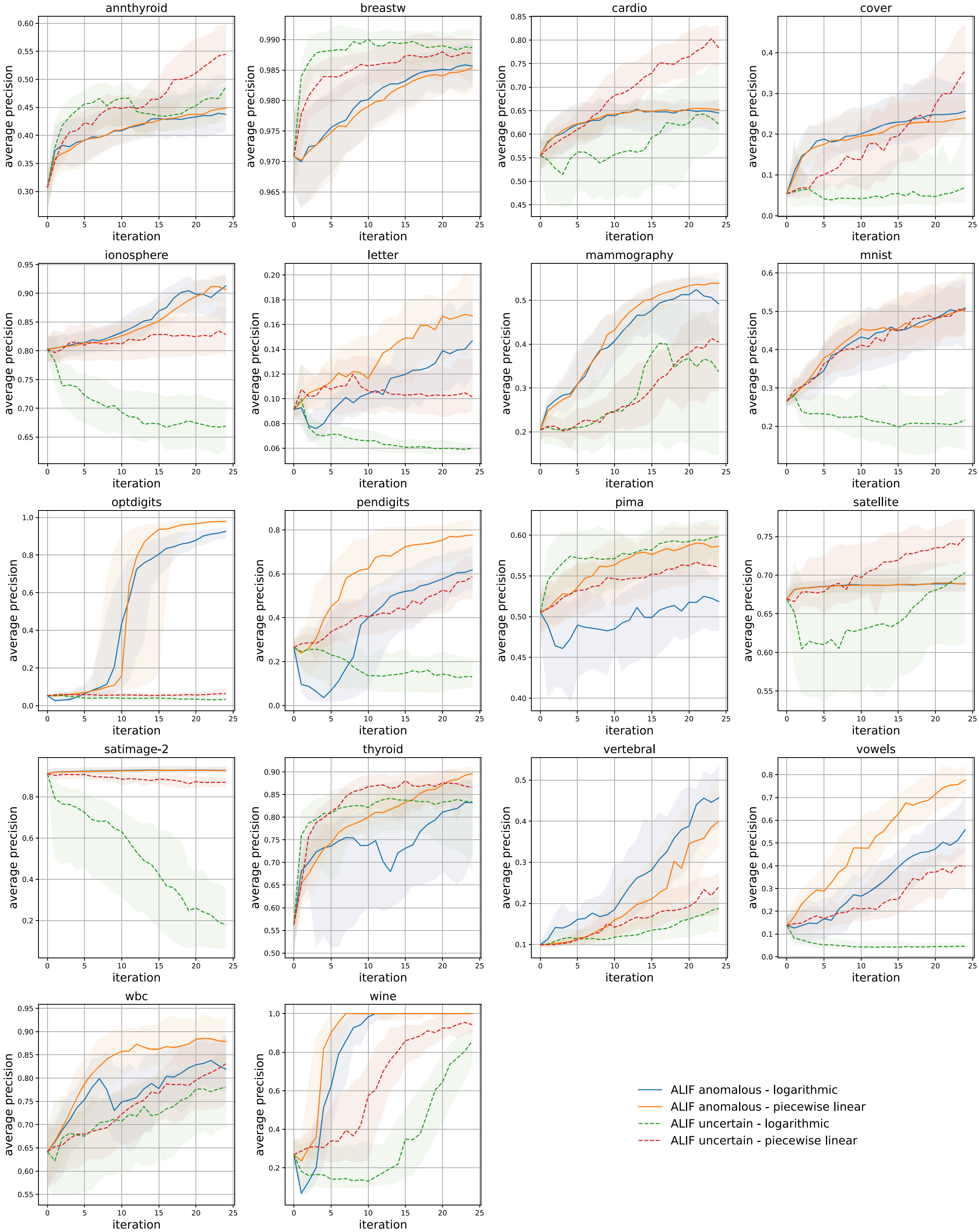

Fig. 6. Performance of the four combinations of the proposed strategies with the datasets described in Table 2. As a general result, it can be noticed that querying the most anomalous point represents the most appropriate choice, overall leading to quicker improvements of the performance. When it comes to the update strategies, both the linear and the logarithmic depths seem to represent a reasonable choice. However, the linear depth appears to moderately be more stable.

Table 3
Summary of the obtained results. The reported performances are the mean performance along the 25 iterations and 50 repetitions of the algorithm. As ν-SSVM is not able to actively query new points, in this comparison ν-SSVM received the labels from ALIF with the most anomalous query strategies, this means that they received the same points and the differences are only due to the different detection model.

			anom-log		anom-lin		unc-log		unc-lin	
	IF-AAD	ν-SSVM	RF	ALIF	RF	ALIF	RF	ALIF	RF	ALIF
annthyroid	0.34	0.18	0.11	0.41	0.11	0.41	0.11	0.44	0.22	**0.45**
breastw	0.93	0.85	0.00	0.98	0.00	0.98	0.28	**0.99**	0.48	0.98
cardio	0.56	0.49	0.15	0.63	0.16	0.63	0.34	0.57	0.49	**0.69**
cover	0.18	*	0.20	0.21	0.18	0.20	0.25	0.09	**0.46**	0.19
ionosphere	0.84	0.78	0.08	**0.85**	0.08	**0.85**	0.47	0.70	0.52	0.82
letter	0.12	0.11	0.08	0.11	0.07	**0.14**	0.06	0.07	0.07	0.11
mammography	0.28	0.20	0.09	0.40	0.10	**0.43**	0.04	0.28	0.18	0.28
mnist	0.36	0.36	0.15	0.41	0.15	**0.43**	0.25	0.24	0.42	0.42
optdigits	0.11	0.04	0.46	0.49	0.38	**0.51**	0.00	0.05	0.00	0.06
pendigits	0.42	0.17	0.28	0.39	0.34	**0.59**	0.10	0.19	0.24	0.42
pima	0.50	0.46	0.44	0.49	0.43	0.56	0.34	**0.57**	0.43	0.54
satellite	0.65	0.47	0.01	0.69	0.01	0.68	0.42	0.64	0.49	**0.70**
satimage-2	0.92	**0.94**	0.00	0.93	0.00	0.93	0.25	0.48	0.53	0.88
thyroid	0.66	0.41	0.37	0.69	0.33	**0.80**	0.10	0.73	0.33	**0.80**
vertebral	0.14	0.12	0.16	**0.27**	0.12	0.22	0.17	0.14	0.22	0.17
vowels	0.29	0.20	0.29	0.33	0.30	**0.51**	0.07	0.06	0.18	0.27
wbc	0.71	0.69	0.68	0.76	0.68	**0.82**	0.11	0.69	0.21	0.73
wine	0.69	0.57	0.73	0.80	0.75	**0.85**	0.28	0.38	0.47	0.64

ALIF consistently beats the other tested approached on all the datasets, in particular the combination of the most anomalous query policy and the piece-wise linear leaf update. (*) Due to the high computational complexity of ν-SSVM it was not feasible to test the *cover* dataset.

fact, this is the query strategy employed in the AAD paradigm. For these reasons, we will use the combination "most anomalous - piecewise linear" for comparison to the other baseline and state-of-the-art competitor.

As our approach ALIF essentially relies on designing an active learning-based modification of the classical Isolation Forest, we decided to compare it mainly with other tree-based models: a fully supervised model, i.e. the Random Forest, and a weakly supervised model, the Isolation Forest - Active Anomaly Detection (IF-AAD). In addition, a quite different model that was compared with ALIF is ν -SSVM [25]. This semi-supervised method does not rely on trees, but it is instead based on support vector machines. The RF represents the baseline since it is the most well-known but simple classification approach; unfortunately, it requires samples from both the classes inlier/outlier and it is computationally expensive as every time the user labels a new data point the forest is retrained. As RF does not have a default method for querying new points, in the following comparison the points given to RF for training are the points queried by ALIF. Similarly, to incorporate new information into ν-SSVM, the entire model needs to be re-trained. This step involves solving a large quadratic programming optimisation problem that scales poorly with the size of the data set. For this reason, this method is not feasible for extremely large datasets, and we were unable to simulate the results for the *cover* dataset. As for RF, ν-SSVM does not provide a natural way to query new points, and therefore, for the comparison the model was trained using the same labels obtained by ALIF queries. On the other hand, IF-AAD is an active semi-supervised model that does not need both class labels and is available online.[3]

Fig. 7 and Table 3 show the benchmark results: it is evident that ALIF obtains the finest results with a very little number of labels, generally outperforming IF-AAD and ν-SSVM. Due to the strong imbalance between inliers and outliers, RF receives quite late both labels from the two classes, leading to poor detection performances: in *breastw*, *satellite*, and *satimage-2* the querying process never got them over 25 iterations and 50 independent repetitions.

5. Conclusions

In this paper a modification of the unsupervised Isolation Forest named ALIF has been suggested to improve the performance of the standard algorithm. Due to the lack of fully labelled datasets, Anomaly Detection is often performed by means of unsupervised models. However, as anomalies heavily depend on the context and are very domain specific, unsupervised models may be unable to detect them because of different definition of anomaly. To solve this problem, we propose a model that starting from an unsupervised model, iteratively tunes it towards the user-definition of anomaly. This allows to enhance the detection performances, avoiding the need to fully label the training dataset and keeping as low as possible the number of required labels. Indeed, this approach takes advantage of the Active Learning framework where the model is able to query the user and select the most interesting samples to label. ALIF relies on two important and interconnected steps: the query policy that selects the point to be labelled, and the model update policy that allows the model to actually learn from the new query. Comparing ALIF with other methods, it turns out that it also generally learns the correct anomaly definition much faster, when the labelling effort needs to be kept low.

[3] https://github.com/shubhomoydas/ad_examples.

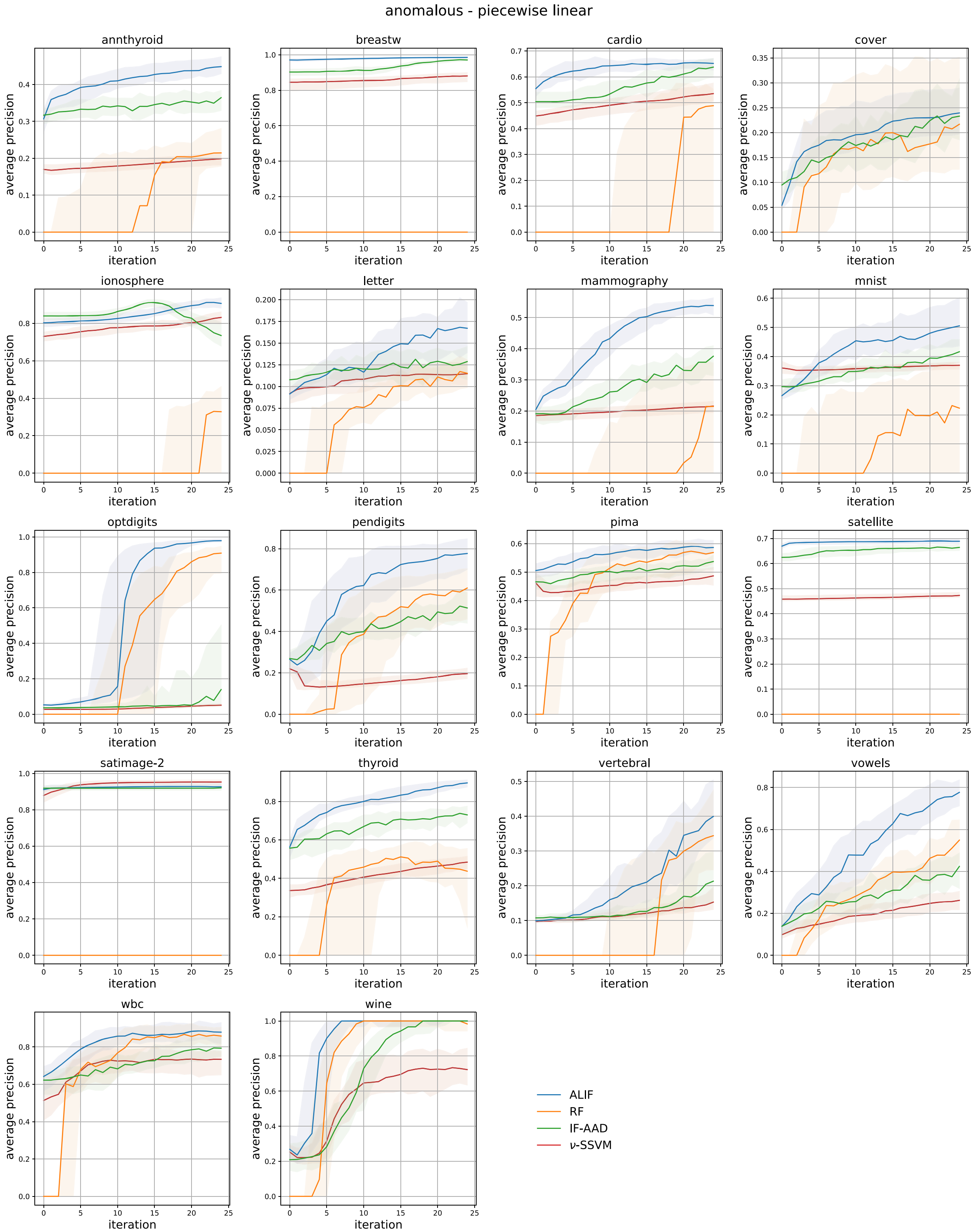

Fig. 7. Comparison of most anomalous - piece-wise linear ALIF with IF-AAD, ν-SSVM and RF. It can be observed that, in general our method represents the best course of action, having the highest performance score usually with a very small amount of labelled data. Due to the computational complexity of ν-SSVM it was not feasible to test the *cover* dataset.

The current study proposed an algorithm to query new points and accordingly update the detection model: the approach was validated on both synthetic and real datasets.

A promising application of the proposed algorithm is to reduce the effects of a possible concept drift: if the concept of anomaly changes, this method could help to adapt the detector in a smooth way. This may happen for example when conditions that the user perceived as anomalous become, with the passing of time, no more worth to be monitored. However, when *data* drift occurs, i.e. when the distribution of the features dynamically changes, this method is not directly equipped with mechanisms to cope with such scenarios. To solve this issue, future works might combine the intuitions discussed in this paper with other tree structures that can learn in a continuous way like Mondrian Forests [23,29].

Moreover, we note that our approach, ALIF, relies on two key assumptions: (i) users or human operators can provide labels to the model, and (ii) these labels are accurate. Assumption (i) is generally valid in decision support systems, while assumption (ii), as previously discussed, is reasonable but not always true: experts, though typically involved, can make errors or may lack complete visibility into complex system states. Therefore, in future work, we will modify our approach to relax assumption (ii), creating a broader framework that accommodates the possibility of erroneous feedback to a certain extent.

Another aspect that is currently under investigation is the development of a theoretically grounded framework to explain the effectiveness of ALIF. Moreover, it seems promising to investigate the different query and updating strategies: the information of the number of data points in each leaf might be included in the algorithm, to estimate the uncertainty on the leaf depth correction. Hybrid query strategies that use a combination of the previously described, might lead to even better results.

CRediT authorship contribution statement

Elisa Marcelli: Conceptualization, Data curation, Formal analysis, Investigation, Methodology, Software, Validation, Writing – original draft. **Tommaso Barbariol:** Conceptualization, Data curation, Formal analysis, Investigation, Methodology, Software, Writing – original draft. **Davide Sartor:** Conceptualization, Data curation, Formal analysis, Investigation, Methodology, Software, Validation, Visualization, Writing – review & editing. **Gian Antonio Susto:** Conceptualization, Formal analysis, Funding acquisition, Investigation, Methodology, Project administration, Resources, Supervision, Validation, Writing – original draft, Writing – review & editing.

Declaration of competing interest

The authors declare the following financial interests/personal relationships which may be considered as potential competing interests:

Gian Antonio Susto reports financial support was provided by European Union.

Data availability

Data is already publicly available.

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Appendix A. Supplementary visualisations

The main focus of this paper is to introduce ALIF, describing its motivations, design choices, limitations and showcasing its performance compared to other active learning anomaly detection approaches. In order to keep the discussion concise, graphs in the main corpus showcase mainly the average precision evolution as a function of labelled instances. However, average precision, as any single statistic, cannot fully capture the model performance. One could argue that other metrics such as F1 score could produce significantly different results. In this appendix we compare directly the two, showcasing how this is in fact not the case. Moreover, in order to provide a more detailed picture of the performance evolution, we also include a query by query snapshot of the precision-recall curve evolution as well as the optimal F1 score achieved by ALIF at the beginning and end of the querying process (see Figs. A.8 and A.9).

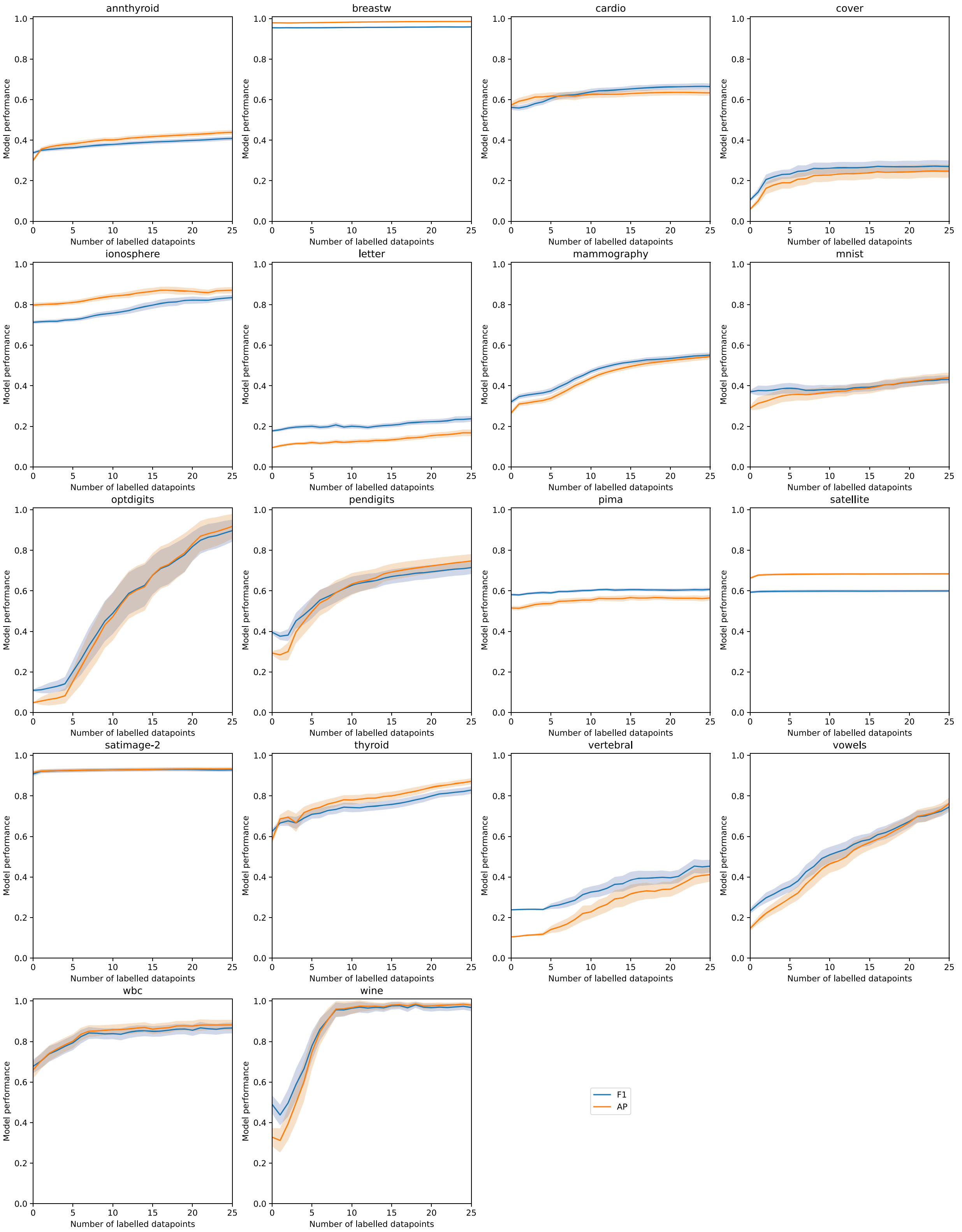

Fig. A.8. Comparison between summarising statistics evolution for ALIF. The plots show how AP and maximum F1 evolve as a function of number of queries.

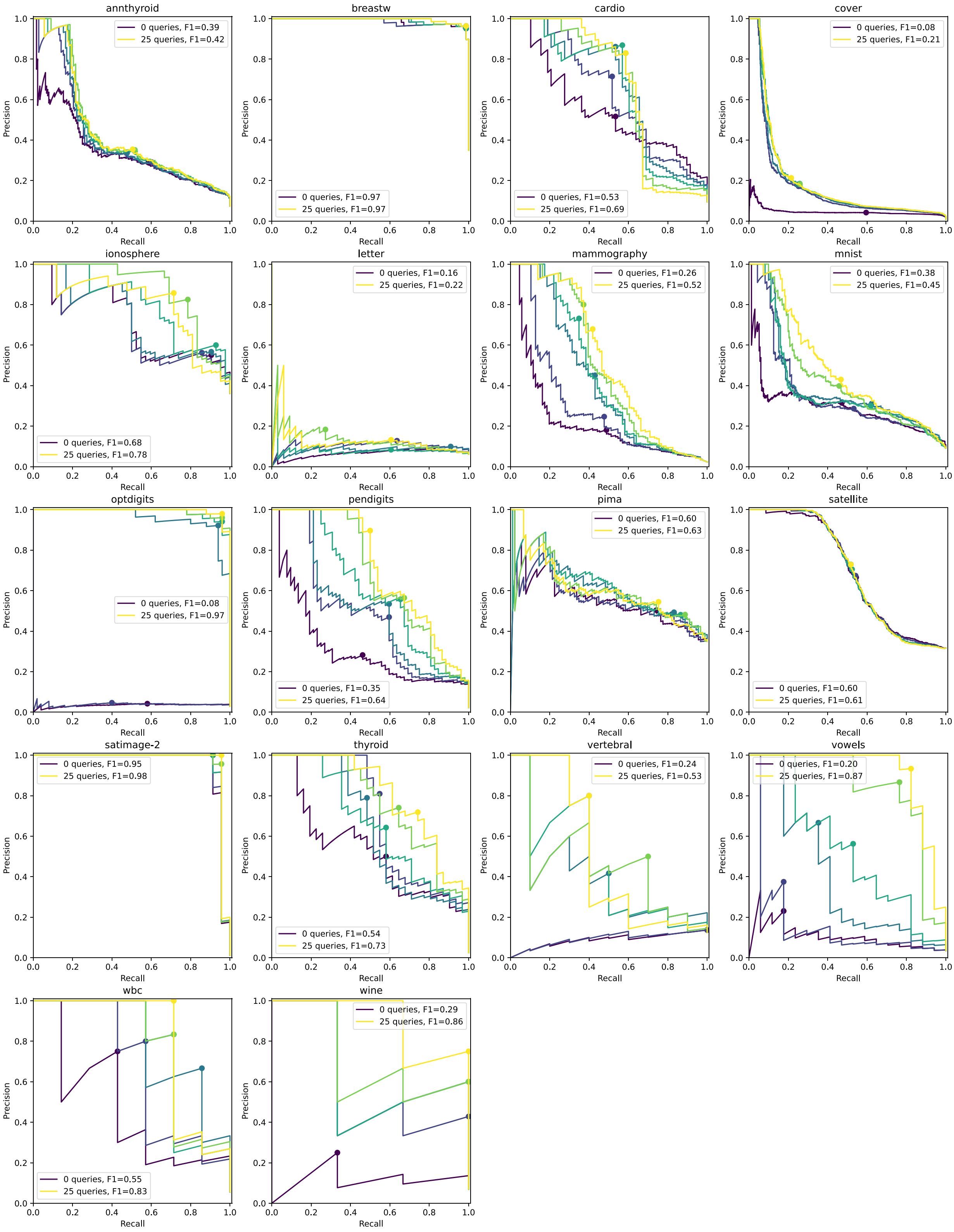

Fig. A.9. Detailed performance evolution of ALIF (one random run). The line plots show the precision recall curve colour coded with the number of queries performed. Solid dots highlight the optimal threshold maximising the F1 score.

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